

Algebraic Number Theory

Adam Kelly (ak2316@cam.ac.uk)

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1 Algebraic Numbers

Definition 1.1. Let $\alpha \in \mathbb{C}$. We say that α is an **algebraic number** with $\alpha \in \overline{\mathbb{Q}}$ if it is the root of a monic polynomial with rational coefficients. If this polynomial has integer coefficients, we say that α is a **algebraic integer** with $\alpha \in \overline{\mathbb{Z}}$.

Proposition 1.2. *The only rational algebraic integers are regular integers.*

Proof. Clearly regular integers are algebraic integers, so we just check the other direction. Let p/q be the root of a monic polynomial $f = \sum_{i=0}^n a_i X^i$ with $a_i \in \mathbb{Z}$ and p, q coprime. Then since $f(p/q) = 0$, we have $q^n f(p/q) = \sum_{i=0}^n a_i p^i q^{n-i}$, and so $q^n f(p/q) \equiv p^n \equiv 0 \pmod{q}$, and since p and q are coprime, we must have $q = \pm 1$, and thus p/q is an integer. $\square$